

Operations/Work/Job Activity covered by this assessment: Plasma

Required Personal Protective Equipment (PPE) :**Safety glasses must be worn at times.****Long and loose hair must be contained.****Appropriate footwear with substantial uppers must be worn.****Close fitting/protective clothing to cover arms and legs must be worn.****Respiratory protection devices required.****Oil free leather gloves and spats must be worn.****Hearing protection must be used when using this machine.****Loading & Unloading Check:**

1. Ensure no slip/trip is present in work pass and walk ways.
2. Ensure that work area is clean and clear of grease, oil and any flammable materials.
3. Refer Work Instructions PSP/WI/14 while handling.

Operations Procedure:

1. Refer Pre-Inspection Work Instructions PSP/WI/12 to pre-inspect the Component.
2. Refer Final Blasting Work Instructions PSP/WI/04 for preparing surface. The surface preparation will be as per Sa3 specs.
3. Mask with High Temperature Adhesive Tape/Metal Shim/ to protect unwanted area from over spraying during Plasma spray coating.
4. Load the Component in the acoustic room and clamp it thoroughly on the chuck with proper packing to prevent clamping marks on the Component.
5. Start Exhaust Fans, Chiller, Water Pump and Capacitor before starting the Plasma Control Unit.
6. Open valve of Argon, Hydrogen Cylinders and Air valve and set the pressure according to the powder parameter mentioned in the Technical Bulletin.
7. Start the Plasma Control unit.
8. Check Argon & Hydrogen Flow & Pressure on the gauges and Flow meters Tubes.
9. Check and Set DC Volts and DC Amperes Reading.
10. Select the required powder w.r.t the Job Card.
11. Pre-heat the powder in the Oven and maintain the temperature to 120 C for two hours.
12. Measure the powder flow rate as per parameter, and also set the flow meter reading.
13. Set the job speed and Gun speed, refer Job Card or instructions will be given by Supervisor/WM.
14. Closed the Acoustic Door and purge the gases from the Plasma control unit, press the

**Works Instruction**

ignition switch to check the ARC.

15. Press the Start switch for ignition. Wait for the gun to ramp up till you get the desire Amp and volts are display on the Plasma Control unit as per the parameter.
16. Pre-heat the skin temperature of the job up to 45°C to 60°C, check with pyrometer. (do not pre-heat the Aluminium components).
17. Select "ON" Powder feeder and wait till the powder is stabilize and spray the job layer by layers to get a uniform thickness all over.
18. Make sure that while spraying the job temperature does not exceed 175°C. You can use Air for cooling the job while spraying.
19. Check the dimension to get the desired thickness of the coating mentioned in the job card.
20. After completion of spraying of the component apply recommended sealant. This should be done before the removing the component from the machine.
21. Intermediate dimension should be checked to control dimension as per job card.
22. Cool the job gradually and recheck the dimensions.
23. Fill up the Job Card as per the data Tag it.

Shutdown :

1. Shut the Powder Feeder and wait for the feed lines to clear.
2. Select the Stop switch to Shut down the ignition. Press purge switch to cool the gun hardware.
3. Stop the Exhaust fan, Chiller and water pump.
4. Stop the main switch to shut the Control Unit.

Assessor :		Signature:		Date:	
Approved:		Signature:		Date:	